

Score-based diffusion models via stochastic differential equations

Wenpin Tang¹ and Hanyang Zhao¹

¹*Department of Industrial Engineering and Operations Research, Columbia University,
e-mail: wt2319@columbia.edu; hzy2684@columbia.edu*

Abstract: This is an expository article on the score-based diffusion models, with a particular focus on the formulation via stochastic differential equations (SDE). After a gentle introduction, we discuss the two pillars in the diffusion modeling – sampling and score matching, which encompass the SDE/ODE sampling, score matching efficiency, the consistency models, and reinforcement learning. Short proofs are given to illustrate the main idea of the stated results. The article is primarily a technical introduction to the field, and practitioners may also find some analysis useful in designing new models or algorithms.

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1. Introduction

Diffusion models describe a family of generative models that genuinely create the desired target distribution from noise. Inspired from energy-based modeling [76], [31, 81, 83] formalized the idea of diffusion models, which underpin the recent success in the text-to-image creators such as DALL·E 2 [70] and Stable Diffusion [73], and the text-to-video generators Sora [64], Make-A-Video [75] and Veo [27]. Roughly speaking, *score-based diffusion models* rely on a forward-backward procedure¹:

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¹There is another class of generative models, the *flow-based models*. One such example is the probabilistic flow ODE (see Section 5) in which:

- The forward process transform the signal to a prescribed noise by a deterministic flow: $X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_n \stackrel{d}{=} p_{\text{noise}}(\cdot)$. The noise $p_{\text{noise}}(\cdot)$ is often called the *prior*.
- The backward process reconstructs the signal from noise: $X_n \rightarrow X_{n-1} \rightarrow \cdots \rightarrow X_0 \stackrel{d}{\approx} p_{\text{data}}(\cdot)$.

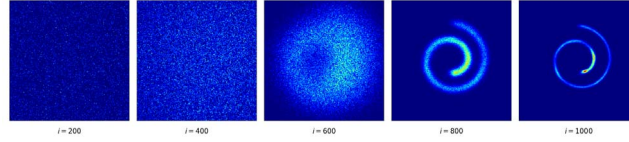
Recent flow matching models (e.g., normalizing flows [57, 66], rectified flows [19, 58], diffusion bridges [15] and consistency models [78, 79]) do not specify a forward process, but instead directly construct a probability path from $p_{\text{noise}}(\cdot)$ to $p_{\text{data}}(\cdot)$.

- Forward deconstruction: starting from the target distribution $X_0 \sim p_{\text{data}}(\cdot)$, the model gradually adds noise to transform the signal into noise $X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_n \stackrel{d}{\approx} p_{\text{noise}}(\cdot)$.
- Backward construction: start with $X_n \sim p_{\text{noise}}(\cdot)$, and reverse the forward process to recover the signal from noise $X_n \rightarrow X_{n-1} \rightarrow \dots \rightarrow X_0 \stackrel{d}{\approx} p_{\text{data}}(\cdot)$.

The forward deconstruction is straightforward. What’s the key in the diffusion models is the backward construction, and the underlying question is how to reverse the forward process. The answer hinges on two pillars: *time reversal* of (Markov) diffusion processes to set the form of the backward process (Section 2), and *score matching* to learn this process (Section 4).



(a) Original Swiss Roll



(b) Swiss Roll generation with 200, 400, 600, 800, 1000 iterations.

FIG 1. *Swiss Roll generation.*

In the original work [31, 77, 81], the forward/backward processes are specified by discrete-time Markov chains; [80, 83] unified the previous models through the lens of stochastic differential equations (SDE). In fact, there is no conceptual distinction between discrete and continuous diffusion models because the diffusion models specified by the SDEs can be regarded as the continuum limits of the discrete models (Section 3), and the discrete diffusion models are obtained from the continuous models by suitable time discretization (Section 6.3). The view is that the SDEs unveil structural properties of the models, whereas the discrete counterparts give practical implementation.

The purpose of this article is to provide a tutorial on the recent theory of score-based diffusion models, mainly from a *continuous perspective* with a *statistical focus*. References on the discrete models will also be given. We sketch the proofs for most stated results, and the assumptions are given only when they are crucial in the analysis. We often use the phrase “under suitable conditions” to avoid less important technical details, and keep the presentation concise and to the point. The paper serves as a gentle introduction to the field, and practitioners may find some analysis useful in designing new models or

algorithms. Since the SDE formulation is adopted, we assume that the readers are familiar with basic stochastic calculus. Øksendal's book [63] provides a user-friendly account for stochastic analysis, and more advanced textbooks are [40, 84]. See also [8, 98] for a literature review and future directions in diffusion models, and [9, 95] for more advanced materials such as diffusion guidance and fine-tuning.

The remainder of the paper is organized as follows. In Section 2, we start with the time reversal formula of diffusion processes, which is the cornerstone of diffusion models. Concrete examples are provided in Section 3. Section 4 is concerned with score matching techniques, another key ingredient of diffusion models. In Section 5, a deterministic sampler – the probability ODE flow is introduced, along with its application to the consistency models. In Section 6, we consider the convergence of diffusion samplers. Additional results on score matching are given in Section 7. Concluding remarks and future directions are summarized in Section 8.

Notations: Below we collect a few notations that will be used throughout.

- For x, y vectors, denote by $x \cdot y$ the inner product between x and y , and $|x|$ the Euclidean (L^2) norm of x .
- For A a matrix (or a vector), $A^\top$ denotes the transpose of A . For A a square matrix, $\text{Tr}(A)$ is the trace of A .
- For $f : \mathbb{R}^d \rightarrow \mathbb{R}$, ∇f is the gradient of f , $\nabla^2 f$ is the Hessian of f , and Δf is the Laplacian of f . For $f = (f_1, \dots, f_n)$ with $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$ its i^{th} column, $(\nabla f)_i^\top = \nabla f_i$ (the Jacobian matrix), and when $n = d$, $\nabla \cdot f = \text{div}(f) = \frac{\partial f_1}{\partial x_1} + \dots + \frac{\partial f_d}{\partial x_d}$ (the divergence). For $f = (f_1, \dots, f_n)$ with $f_i : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $(\nabla \cdot f)_i = \text{div}(f_i)$ for $i = 1, \dots, n$.
- The symbol $a = \mathcal{O}(b)$ means that a/b is bounded, and $a = o(b)$ means that a/b tends to zero as some problem parameter tends to 0 or ∞ .
- For X a random variable, $\mathcal{L}(X)$ denotes the probability distribution of X , and $X \sim p(\cdot)$ means that X has the distribution $p(\cdot)$. The notation $X \stackrel{d}{=} Y$ represents that X and Y have the same distribution.
- $\mathbb{E}X$ and $\text{Var}(X)$ denote the expectation and the variance of X respectively. We use $\widehat{\mathbb{E}}X$ for the empirical average of X , i.e. $\widehat{\mathbb{E}}X = \frac{1}{n} \sum_{i=1}^n X_i$ for $X_1, \dots, X_n$ independent and identically distributed copies of X .
- $\mathcal{N}(\mu, \Sigma)$ denotes Gaussian distribution with mean μ and covariance matrix Σ .
- For P and Q two probability distributions, $d_{TV}(P, Q) = \sup_A |P(A) - Q(A)|$ is the total variation distance between P and Q ; $\text{KL}(P, Q) = \int \log \left(\frac{dP}{dQ} \right) dP$ is the Kullback-Leibler (KL) divergence between P and Q ; $W_2(P, Q) = \left(\inf_{\gamma} \mathbb{E}_{(X, Y) \sim \gamma} |X - Y|^2 \right)^{\frac{1}{2}}$, where the infimum is taken over all couplings γ of P and Q , is the Wasserstein-2 distance between P and Q .

2. Time reversal formula

In this section, we present the time reversal formula, which lays the foundation for score-based diffusion models.

Consider a general (forward) SDE:

$$dX_t = f(t, X_t)dt + g(t, X_t)dB_t, \quad X_0 \sim p_{\text{data}}(\cdot), \quad (1)$$

where $(B_t, t \geq 0)$ is n -dimensional Brownian motion², and $f : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $g : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times n}$ are diffusion parameters. Some conditions on $f(\cdot, \cdot)$ and $g(\cdot, \cdot)$ are required so that the SDE (1) is well-defined. For instance,

- If f and g are Lipschitz and have linear growth in x uniformly in t , then (1) has a strong solution which is pathwise unique.
- If f is bounded, and g is bounded, continuous and strictly elliptic, then (1) has a weak solution which is unique in distribution.

See [40, 63] for background on the well-posedness of SDEs, and [13, Chapter 1] for a review of related results.

For ease of presentation, we assume that X_t has a (suitably smooth) probability density $p(t, \cdot)$. Let $T > 0$ be fixed, and run the SDE (1) until time T to get $X_T \sim p(T, \cdot)$. Now if we start with $p(T, \cdot)$ and run the process X backward for time T , then we can generate copies of $p(0, \cdot) = p_{\text{data}}(\cdot)$. More precisely, consider the time reversal $Y_t := X_{T-t}$ for $0 \leq t \leq T$. Assuming that Y also satisfies an SDE, we can run the backward process:

$$dY_t = \bar{f}(t, Y_t)dt + \bar{g}(t, Y_t)dB_t, \quad Y_0 \sim p(T, \cdot).$$

So we generate the desired $Y_T \sim p_{\text{data}}(\cdot)$ at time T .

As mentioned earlier, the high-level idea of diffusion models is to create the target distribution from noise. This means that the noise should not depend on the target distribution. Thus, we replace the (backward) initialization $Y_0 \sim p(T, \cdot)$ with some noise $p_{\text{noise}}(\cdot)$:

$$dY_t = \bar{f}(t, Y_t)dt + \bar{g}(t, Y_t)dB_t, \quad Y_0 \sim p_{\text{noise}}(\cdot), \quad (2)$$

as an approximation. Two natural questions arise:

1. How do we choose the noise $p_{\text{noise}}(\cdot)$?
2. What are the coefficients $\bar{f}(\cdot, \cdot)$ and $\bar{g}(\cdot, \cdot)$?

For (1), the noise $p_{\text{noise}}(\cdot)$ should be easily sampled. It is commonly derived by decoupling X_0 from the conditional distribution of $(X_T | X_0)$, as we will explain with the examples in Section 3. It is expected that the closer the distributions $p(T, \cdot)$ and $p_{\text{noise}}(\cdot)$ are, the closer the distribution of Y_T sampled from (2) is to $p_{\text{data}}(\cdot)$. For (2), it relies on the following result on the time reversal of SDEs. To simplify the notations, we write

$$a(t, x) := g(t, x)g(t, x)^\top.$$

²Write $B = (B^1, \dots, B^n)$, where B^i 's are independent standard Brownian motions.

Theorem 2.1 (Time reversal formula). [1, 30] Under suitable conditions on $f(\cdot, \cdot)$, $g(\cdot, \cdot)$ and $\{p(t, \cdot)\}_{0 \leq t \leq T}$, we have:

$$\bar{g}(t, x) = g(T-t, x), \quad \bar{f}(t, x) = -f(T-t, x) + \frac{\nabla \cdot (p(T-t, x)a(T-t, x))}{p(T-t, x)}. \quad (3)$$

Proof. We give a heuristic derivation of the formula (3), which consists of identifying the Fokker–Planck equation of $\bar{p}(t, x) := p(T-t, x)$ (the probability density of the time reversal Y). First, the infinitesimal generator of X is $\mathcal{L} := \frac{1}{2}\nabla \cdot a(t, x)\nabla + f_a \cdot \nabla$, where $f_a := f - \frac{1}{2}\nabla \cdot a$. It is known that the density $p(t, x)$ satisfies the Fokker–Planck equation:

$$\frac{\partial}{\partial t}p(t, x) = \mathcal{L}^*p(t, x),$$

where $\mathcal{L}^* := \frac{1}{2}\nabla \cdot a(t, x)\nabla - \nabla \cdot f_a$ is the adjoint operator of $\mathcal{L}$. Thus,

$$\frac{\partial}{\partial t}\bar{p}(t, x) = -\frac{1}{2}\nabla \cdot (a(T-t, x)\nabla\bar{p}(t, x)) + \nabla \cdot (f_a(T-t, x)\bar{p}(t, x)). \quad (4)$$

On the other hand, we expect the generator of Y to be $\bar{\mathcal{L}} := \frac{1}{2}\nabla \cdot \bar{a}(t, x)\nabla + \bar{f}_{\bar{a}} \cdot \nabla$. The Fokker–Planck equation for $\bar{p}(t, x)$ is:

$$\frac{\partial}{\partial t}\bar{p}(t, x) = \frac{1}{2}\nabla \cdot (\bar{a}(t, x)\nabla\bar{p}(t, x)) - \nabla \cdot (\bar{f}_{\bar{a}}(t, x)\bar{p}(t, x)). \quad (5)$$

Comparing (4) and (5), we set $\bar{a}(t, x) = a(T-t, x)$ and then get

$$(f_a(T-t, x) + \bar{f}_{\bar{a}}(t, x))\bar{p}(t, x) = a(T-t, x)\nabla\bar{p}(t, x),$$

which can be rewritten as:

$$(f(T-t, x) + \bar{f}(t, x))\bar{p}(t, x) - \nabla \cdot a(T-t, x)\bar{p}(t, x) = a(T-t, x)\nabla\bar{p}(t, x).$$

This yields the desired result. $\square$

Let's comment on Theorem 2.1. [30, 62] proved the result by assuming that $f(\cdot, \cdot)$ and $g(\cdot, \cdot)$ are globally Lipschitz, and the density $p(t, x)$ satisfies an a priori H^1 bound. The implicit condition on $p(t, x)$ is guaranteed if $\partial_t + \mathcal{L}$ is hypoelliptic, or $\nabla^2 a(t, x)$ is uniformly bounded. These conditions were relaxed in [69], where only the boundedness of $\nabla a(t, x)$ in some L^2 norm is required. In another direction, [22, 23] used an entropy argument to prove the time reversal formula in the case of $n = d$ and $g(t, x) = \sigma I$. This approach was further developed by [6] in connection with optimal transport theory.

By Theorem 2.1, the backward process is:

$$dY_t = (-f(T-t, Y_t) + a(T-t, Y_t)\nabla \log p(T-t, Y_t) + \nabla \cdot a(T-t, Y_t))dt + g(T-t, Y_t)dB_t, \quad Y_0 \sim p_{\text{noise}}(\cdot). \quad (6)$$

Since $f(\cdot, \cdot)$ and $g(\cdot, \cdot)$ are chosen in advance, all but the term $\nabla \log p(T-t, Y_t)$ in (6) are available. So in order to run the backward process (6), we need to learn $\nabla \log p(t, x)$, known as *Stein's score function*. Recently developed score-based generative modeling consists of estimating $\nabla \log p(t, x)$ by function approximations, which will be discussed in Section 4.

3. Examples

We have seen that a diffusion model is specified by the pair $(f(\cdot, \cdot), g(\cdot, \cdot))$. The design of $(f(\cdot, \cdot), g(\cdot, \cdot))$ is important because it determines the quality of data generation. There are two general rules of thumb: easy learning from the forward process, and good sampling from the backward process, which will be clear in the next two sections.

Now let's provide some examples of the diffusion model (1)–(6). Most existing models take the form:

$$n = d \quad \text{and} \quad g(t, x) = g(t)I,$$

where $g(t) \in \mathbb{R}_+$. That is, the model parameter $g(\cdot, \cdot)$ is only time-dependent, rather than (time and) state-dependent. One important reason is that for the SDEs with state-dependent coefficient, it is often not easy to decouple X_0 from the distribution of $(X_T | X_0)$. Hence, it is not clear how to pick $p_{\text{noise}}(\cdot)$ as a proxy to $X_T \sim p(T, \cdot)$, as is the case of geometric Brownian motion.

(a) *Ornstein-Uhlenback (OU) process* [15]:

$$f(t, x) = \theta(\mu - x) \text{ with } \theta > 0, \mu \in \mathbb{R}^d \quad \text{and} \quad g(t) = \sigma > 0. \quad (7)$$

The distribution of $(X_t | X_0 = x)$ is $p(t, \cdot; x) = \mathcal{N}(\mu + (x - \mu)e^{-\theta t}, \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t})I)$. An obvious candidate for the noise is the stationary distribution of the OU process $p_{\text{noise}}(\cdot) = \mathcal{N}(\mu, \frac{\sigma^2}{2\theta}I)$. The backward process is specified to:

$$dY_t = (\theta(Y_t - \mu) + \sigma^2 \nabla \log p(T - t, Y_t)) dt + g dB_t, \quad Y_0 \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{2\theta}I\right). \quad (8)$$

More generally, consider the overdamped Langevin process:

$$f(t, x) = -\nabla U(x) \quad \text{and} \quad g(t) = \sigma > 0, \quad (9)$$

with a suitable landscape $U : \mathbb{R}^d \rightarrow \mathbb{R}$. The OU process corresponds to the choice of $U(x) = \frac{\theta}{2}|x - \mu|^2$. The backward process is then:

$$dY_t = (\nabla U(Y_t) + \sigma^2 \nabla \log p(T - t, Y_t)) dt + \sigma dB_t, \quad Y_0 \propto \exp\left(-\frac{2U(\cdot)}{\sigma^2}\right). \quad (10)$$

(b) *Variance exploding (VE) SDE* [83]: This is the continuum limit of *score matching with Langevin dynamics* (SMLD) [81]. The idea of SMLD is to use N noise scales $\sigma_0 < \sigma_1 < \dots < \sigma_{N-1}$, and run the (forward) Markov chain:

$$x_i = x_{i-1} + \sqrt{\sigma_i^2 - \sigma_{i-1}^2} z_{i-1}, \quad 1 \leq i \leq N,$$

where $z_0, \dots, z_{N-1}$ are independent and identically distributed $\mathcal{N}(0, I)$. By taking $\Delta t = \frac{T}{N} \rightarrow 0$, $\sigma(i\Delta t) = \sigma_i$, $x_{i\Delta t} = x_i$ and $z_{i\Delta t} = z_i$, we get $x_{t+\Delta t} = x_t + \sqrt{\sigma^2(t + \Delta t) - \sigma^2(t)} z_t \approx x_t + \sqrt{\frac{d\sigma^2(t)}{dt} \Delta t} z_t$. That is,

$$dX_t = \sqrt{\frac{d\sigma^2(t)}{dt}} dB_t, \quad 0 \leq t \leq T.$$

This implies that $f(t, x) = 0$ and $g(t) = \sqrt{\frac{d\sigma^2(t)}{dt}}$. The noise scales are typically set to be a geometric sequence $\sigma(t) = \sigma_{\min} \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)^{\frac{t}{T}}$, with $\sigma_{\min} \ll \sigma_{\max}$. Thus,

$$f(t, x) = 0 \quad \text{and} \quad g(t) = \sigma_{\min} \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)^{\frac{t}{T}} \sqrt{\frac{2}{T} \log \frac{\sigma_{\max}}{\sigma_{\min}}}. \quad (11)$$

Here $X_t = X_0 + \int_0^t g(s) dB_s$ is the Paley-Wiener integral. The distribution of $(X_t | X_0 = x)$ is:

$$p(t, \cdot; x) = \mathcal{N} \left(x, \left(\int_0^t g^2(s) ds \right) I \right) = \mathcal{N} \left(x, \sigma_{\min}^2 \left(\left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)^{\frac{2t}{T}} - 1 \right) I \right). \quad (12)$$

The name “variance exploding” comes from the fact that $\text{Var}(X_0) \ll \text{Var}(X_T)$ because $\sigma_{\min} \ll \sigma_{\max}$. Note that the forward process X does not have a stationary distribution, but we can decouple x from $p(T, \cdot) = \mathcal{N}(x, (\sigma_{\max}^2 - \sigma_{\min}^2)I)$ to get

$$p_{\text{noise}}(\cdot) = \mathcal{N}(0, (\sigma_{\max}^2 - \sigma_{\min}^2)I).$$

The backward process is:

$$dY_t = g^2(T-t) \nabla \log p(T-t, Y_t) + g(T-t) dB_t, \quad Y_0 \sim \mathcal{N}(0, (\sigma_{\max}^2 - \sigma_{\min}^2)I), \quad (13)$$

where $g(t)$ is defined by (11).

As we will see in Section 5, it is more convenient to use $\sigma(t) = t$ and $g(t) = \sqrt{2t}$ (suggested by [41] as a base model), with $p_{\text{noise}}(\cdot) = \mathcal{N}(0, T^2 I)$ as an alternative parametrization.

(c) *Variance preserving (VP) SDE* [83]: This is the continuum limit of *denoising diffusion probabilistic models* (DDPM) [31]. DDPM uses N noise scales $\beta_1 < \beta_2 < \dots < \beta_N$, and runs the (forward) Markov chain:

$$x_i = \sqrt{1 - \beta_i} x_{i-1} + \sqrt{\beta_i} z_i, \quad 1 \leq i \leq N.$$

Similarly, by taking the limit $\Delta t = \frac{T}{N} \rightarrow 0$ with $\beta(i\Delta t) = N\beta_i/T$, $x_{i\Delta t} = x_i$ and $z_{i\Delta t} = z_i$, we get $x_{t+\Delta t} = \sqrt{1 - \beta(t)\Delta t} x_t + \sqrt{\beta(t)\Delta t} z_t \approx x_t - \frac{1}{2}\beta(t)x_t\Delta t + \sqrt{\beta(t)\Delta t} z_t$. This leads to:

$$dX_t = -\frac{1}{2}\beta(t)X_t dt + \sqrt{\beta(t)} dB_t, \quad 0 \leq t \leq T.$$

We have:

$$f(t, x) = -\frac{1}{2}\beta(t)x \quad \text{and} \quad g(t) = \sqrt{\beta(t)}. \quad (14)$$

The noise scales of DDPM are typically an arithmetic sequence:

$$\beta(t) = \beta_{\min} + \frac{t}{T}(\beta_{\max} - \beta_{\min}), \quad \text{with } \beta_{\min} \ll \beta_{\max}, \quad (15)$$

(and $\beta_{\min}, \beta_{\max}$ are scaled to the order N/T .) By applying Itô's formula to $e^{\frac{1}{2} \int_0^t \beta(s) ds} X_t$, we get the distribution of $(X_t | X_0 = x)$:

$$p(t, \cdot; x) = \mathcal{N}\left(e^{-\frac{1}{2} \int_0^t \beta(s) ds} x, (1 - e^{-\int_0^t \beta(s) ds}) I\right). \quad (16)$$

Thus, $p(T, \cdot; x) = \mathcal{N}(e^{-\frac{T}{4}(\beta_{\max} + \beta_{\min})} x, (1 - e^{-\frac{T}{2}(\beta_{\max} + \beta_{\min})}) I)$, which is close to $\mathcal{N}(0, I)$ if $\beta_{\max}$, or T is set to be large. This justifies the name “variance preserving”, and we can set $p_{\text{noise}} = \mathcal{N}(0, I)$. The backward process is:

$$dY_t = \left(\frac{1}{2} \beta(T-t) Y_t + \beta(T-t) \nabla \log p(T-t, Y_t) \right) dt + \sqrt{\beta(T-t)} dB_t, \quad (17)$$

$$Y_0 \sim \mathcal{N}(0, I),$$

where $\beta(t)$ is defined by (15).

Note that the choice of $\beta(t) = \sigma^2$ leads to the OU process with $f(t, x) = -\frac{1}{2} \sigma^2 x$ and $g(t) = \sigma$. In fact, the VP SDE is also referred to as the OU process in the literature [11, 47, 48]. Here we emphasize the difference between the OU process, and the VP SDE with the (typical) choice (15) for $\beta(t)$.

(d) *Sub-variance preserving (subVP) SDE* [83]:

$$f(t, x) = -\frac{1}{2} \beta(t) x \quad \text{and} \quad g(t) = \sqrt{\beta(t)(1 - e^{-2 \int_0^t \beta(s) ds})}, \quad (18)$$

where $\beta(t)$ is defined by (15). The same reasoning as in (c) shows that the distribution of $(X_t | X_0 = x)$ is:

$$\begin{aligned} p(t, \cdot; x) &= \mathcal{N}\left(e^{-\frac{1}{2} \int_0^t \beta(s) ds} x, (1 - e^{-\int_0^t \beta(s) ds})^2 I\right) \\ &= \mathcal{N}\left(e^{-\frac{t^2}{4T}(\beta_{\max} - \beta_{\min}) - \frac{t}{2} \beta_{\min}} x, (1 - e^{-\frac{t^2}{2T}(\beta_{\max} - \beta_{\min}) - t \beta_{\min}})^2 I\right). \end{aligned} \quad (19)$$

Note that $\text{Var}_{\text{sub-VP}}(X_t) \leq \text{Var}_{\text{VP}}(X_t)$, hence the name “sub-VP”, and $p_{\text{noise}}(\cdot) = \mathcal{N}(0, I)$. Set

$$\gamma(t) := e^{-2 \int_0^t \beta(s) ds} = e^{-\frac{t^2}{T}(\beta_{\max} - \beta_{\min}) - 2t \beta_{\min}},$$

so $g(t) = \sqrt{\beta(t)(1 - \gamma(t))}$. The backward process is:

$$\begin{aligned} dY_t &= \left(\frac{1}{2} \beta(T-t) Y_t + \beta(T-t)(1 - \gamma(T-t)) \nabla \log p(T-t, Y_t) \right) dt \\ &\quad + \sqrt{\beta(T-t)(1 - \gamma(T-t))} dB_t, \quad Y_0 \sim \mathcal{N}(0, I). \end{aligned} \quad (20)$$

(e) *Contractive diffusion probabilistic models (CDPM)* [89]: The idea of CDPM is to force contraction on the backward process to narrow the score matching errors (at the cost of possible noise approximation bias). A practical criterion is:

$$(x - x') \cdot (f(t, x) - f(t, x')) \geq r_f(t) |x - x'|^2, \quad \text{with} \quad \inf_{t \in [0, T]} r_f(t) > 0. \quad (21)$$

For instance, we consider *contractive variance preserving (CVP) SDE*:

$$f(t, x) = \frac{1}{2}\beta(t)x \quad \text{and} \quad g(t) = \sqrt{\beta(t)}, \quad (22)$$

where $\beta(t)$ is defined by (15). Similar to (c), we set

$$p_{\text{noise}}(\cdot) = \mathcal{N}\left(0, (e^{\frac{T}{2}(\beta_{\max} + \beta_{\min})} - 1)I\right).$$

The backward process is:

$$\begin{aligned} dY_t = & \left(-\frac{1}{2}\beta(T-t)Y_t + \beta(T-t)\nabla \log p(T-t, Y_t) \right) dt \\ & + \sqrt{\beta(T-t)}dB_t, \quad Y_0 \sim \mathcal{N}\left(0, (e^{\frac{T}{2}(\beta_{\max} + \beta_{\min})} - 1)I\right). \end{aligned} \quad (23)$$

Among these examples, VE and VP SDEs are the most widely used models (e.g., Stable Diffusion uses VP, and consistency models rely on VE.) CDPM achieves slightly better performance on certain tasks, but needs to fine-tune its hyperparameters to trade off noise approximation bias and score matching errors. In each of the above models, one can estimate the corresponding score function $\nabla \log p(t, x)$ for backward sampling – this is inconvenient because score matching often requires large computational cost. Nevertheless, there is a reparametrization trick via the probability flow ODE that allows to sample from different models with a single score function of VE. This will be discussed in Section 5.1.

4. Score matching techniques

In Section 2, we have seen that the main obstacle for backward sampling (6) is the unknown score function $\nabla \log p(t, x)$. This section reviews the recently developed score-based generative modeling [31, 81, 83], whose goal is to estimate the score function $\nabla \log p(t, x)$ by a family of functions $\{s_\theta(t, x)\}_\theta$ (e.g., kernels and neural nets). This technique is referred to as *score matching*. With the (true) score function $\nabla \log p(t, x)$ being replaced with the score matching function $s_\theta(t, x)$, the backward process (6) becomes:

$$\begin{aligned} dY_t = & (-f(T-t, Y_t) + a(T-t, Y_t)s_\theta(T-t, Y_t) + \nabla \cdot a(T-t, Y_t))dt \\ & + g(T-t, Y_t)dB_t, \quad Y_0 \sim p_{\text{noise}}(\cdot). \end{aligned} \quad (24)$$

In the case of $g(t, x) = g(t)I$, it simplifies to:

$$\begin{aligned} dY_t = & (-f(T-t, Y_t) + g^2(T-t)s_\theta(T-t, Y_t))dt + g(T-t)dB_t, \\ & Y_0 \sim p_{\text{noise}}(\cdot). \end{aligned} \quad (25)$$

The plan of this section is as follows. In Section 4.1, we present the general score matching technique. The two (popularly used) scalable score matching methods – *sliced score matching* and *denoising score matching* will be discussed in Sections 4.2 and 4.3.

4.1. Score matching

Recall that $\{s_\theta(t, x)\}_\theta$ is a family of functions on $\mathbb{R}_+ \times \mathbb{R}^d$ parametrized by θ , which are used to approximate the score function $\nabla \log p(t, x)$. Fix time t , the goal is to solve the stochastic optimization problem:

$$\min_{\theta} \mathcal{J}_{\text{ESM}}(\theta) := \mathbb{E}_{p(t, \cdot)} |s_\theta(t, X) - \nabla \log p(t, X)|^2, \quad (26)$$

where $\mathbb{E}_{p(t, \cdot)}$ denotes the expectation taken over $X \sim p(t, \cdot)$. But the problem (26), known as the *explicit score matching* (ESM), is far-fetched because the score $\nabla \log p(t, X)$ on the right side is not available.

Interestingly, this problem has been studied in the context of estimating statistical models with unknown normalizing constant. (In fact, if $p(\cdot)$ is a Gibbs measure, then its score $\nabla \log p(\cdot)$ does not depend on the normalizing constant.) The following result shows that the score matching problem (26) can be recast into a feasible stochastic optimization with no $\nabla \log p(t, X)$ -term, referred to as the *implicit score matching* (ISM).

Theorem 4.1 (Implicit score matching). *[36] Let*

$$\mathcal{J}_{\text{ISM}}(\theta) := \mathbb{E}_{p(t, \cdot)} [|s_\theta(t, X)|^2 + 2 \nabla \cdot s_\theta(t, X)]. \quad (27)$$

Under suitable conditions on s_θ , we have $\mathcal{J}_{\text{ISM}}(\theta) = \mathcal{J}_{\text{ESM}}(\theta) + C$ for some C independent of θ . Consequently, the minimum point of $\mathcal{J}_{\text{ISM}}$ and that of $\mathcal{J}_{\text{ESM}}$ coincide.

Proof. We have

$$\begin{aligned} \mathcal{J}_{\text{ESM}}(\theta) &= \mathbb{E}_{p(t, \cdot)} [|\nabla \log p(t, X)|^2] \\ &= \mathbb{E}_{p(t, \cdot)} [|s_\theta(t, X)|^2] - 2 \mathbb{E}_{p(t, \cdot)} [s_\theta(t, X)^\top \nabla \log p(t, X)] \\ &= \mathbb{E}_{p(t, \cdot)} [|s_\theta(t, X)|^2] - 2 \int s_\theta(t, x)^\top \nabla p(t, x) dx \\ &= \mathbb{E}_{p(t, \cdot)} [|s_\theta(t, X)|^2] + 2 \int \nabla \cdot s_\theta(t, x) p(t, x) dx \\ &= \mathbb{E}_{p(t, \cdot)} [|s_\theta(t, X)|^2 + 2 \nabla \cdot s_\theta(t, X)] = \mathcal{J}_{\text{ISM}}(\theta), \end{aligned}$$

where we use the divergence theorem in the third equation. $\square$

Clearly, the implicit score matching problem (27) can be solved by stochastic optimization tools, e.g., stochastic gradient descent (SGD). Here the distribution $p(t, \cdot)$ is sampled by first sampling a random data point from $p_{\text{data}}(\cdot)$, followed by the conditional distribution of $(X_t | X_0)$. This requires the distribution of $(X_t | X_0)$ be easy to sample – the easy learning criterion for diffusion models. Notably, the distribution of $(X_t | X_0)$ is Gaussian for all the examples in Section 3.

In practice, a time-weighted version of the problem (26) is considered:

$$\begin{aligned} \min_{\theta} \tilde{\mathcal{J}}_{\text{ESM}}(\theta) &:= \mathbb{E}_{t \in \mathcal{U}(0, T)} \mathbb{E}_{p(t, \cdot)} [\lambda(t) |s_{\theta}(t, X) - \nabla \log p(t, X)|^2] \\ &= \frac{1}{T} \int_0^T \mathbb{E}_{p(t, \cdot)} [\lambda(t) |s_{\theta}(t, X) - \nabla \log p(t, X)|^2] dt, \end{aligned} \quad (28)$$

where $\mathcal{U}(0, T)$ denotes the uniform distribution over $[0, T]$, and $\lambda : \mathbb{R} \rightarrow \mathbb{R}_+$ is a weight function. The corresponding implicit score matching problem is:

$$\min_{\theta} \tilde{\mathcal{J}}_{\text{ISM}}(\theta) = \mathbb{E}_{t \in \mathcal{U}(0, T)} \mathbb{E}_{p(t, \cdot)} [\lambda(t) (|s_{\theta}(t, X)|^2 + 2 \nabla \cdot s_{\theta}(t, X))]. \quad (29)$$

Note, however, that the problem (27) or (29) can still be computationally expensive when the dimension d is large. For instance, if we use a neural net as the function class of $s_{\theta}(t, x)$, we need to perform d times backpropagation of all the parameters to compute $\nabla \cdot s_{\theta}(t, x) = \text{Tr}(\nabla s_{\theta}(t, x))$. This means that the computation of the derivatives scales linearly with the dimension, hence making the gradient descent methods inefficient to solve the score matching problem with respect to high dimensional data. Two alternative approaches are commonly used to deal with the scalability issue, which will be the focus of the next subsections.

4.2. Sliced score matching

The burden of computation comes from the term $\nabla \cdot s_{\theta}(t, x)$. One clever idea proposed by [82] is to tackle this term by random projections. It relies on the key observation:

$$\nabla \cdot s_{\theta}(t, x) = \mathbb{E}_{v \sim \mathcal{N}(0, I)} [v^{\top} \nabla s_{\theta}(t, x) v], \quad (30)$$

where $\nabla s_{\theta}(t, x) \in \mathbb{R}^{d \times d}$ is the Jacobian matrix of $s_{\theta}(t, x)$. The (implicit) score matching problem (29) can then be rewritten as:

$$\min_{\theta} \tilde{\mathcal{J}}_{\text{SSM}}(\theta) = \mathbb{E}_{t \in \mathcal{U}(0, T)} \mathbb{E}_{v \sim \mathcal{N}(0, I)} \mathbb{E}_{p(t, \cdot)} [\lambda(t) (|s_{\theta}(t, X)|^2 + 2 v^{\top} \nabla (v^{\top} s_{\theta}(t, x)))], \quad (31)$$

which is referred to as the *sliced score matching* (SSM). Since $\tilde{\mathcal{J}}_{\text{SSM}}(\theta) = \tilde{\mathcal{J}}_{\text{ISM}}(\theta)$ for all θ , we have the following result.

Theorem 4.2 (Sliced score matching). [82] *Under suitable conditions on s_{θ} , we have $\tilde{\mathcal{J}}_{\text{SSM}}(\theta) = \tilde{\mathcal{J}}_{\text{ESM}}(\theta) + C$ for some C independent of θ . Consequently, the minimum point of $\tilde{\mathcal{J}}_{\text{SSM}}(\theta)$ and that of $\tilde{\mathcal{J}}_{\text{ESM}}(\theta)$ coincide.*

Note that for a single fixed v , it only requires one-time backpropagation because the term $v^{\top} s_{\theta}(t, x)$ can be regarded as adding a layer of the inner product between v and $s_{\theta}(t, x)$. To get the expectation $\mathbb{E}_{v \sim \mathcal{N}(0, I)}$ in (31), we pick m samples of $v_i \sim \mathcal{N}(0, I)$, $1 \leq i \leq m$, compute the objective for each v_i , and then take the average. So it requires m times backpropagation. Typically, m is set to be small ($m \ll d$), and empirical studies show that $m = 1$ is often good enough.

4.3. Denoising score matching

The second approach relies on conditioning X_t on $X_0 \sim p_{data}(\cdot)$, known as denoising score matching (DSM) [36, 92]. Let's go back to the ESM problem (28), and it is equivalent to the following DSM problem:

$$\begin{aligned} \tilde{\mathcal{J}}_{\text{DSM}}(\theta) &= \mathbb{E}_{t \sim \mathcal{U}(0, T)} \left\{ \lambda(t) \mathbb{E}_{X_0 \sim p_{data}(\cdot)} \mathbb{E}_{X_t | X_0} \left[|s_\theta(t, X_t) - \nabla \log p(t, X_t | X_0)|^2 \right] \right\}, \end{aligned} \quad (32)$$

where the gradient $\nabla \log p(t, X_t | X_0)$ is with respect to X_t .

Theorem 4.3 (Denoising score matching). [92] *Under suitable conditions on s_θ , we have $\tilde{\mathcal{J}}_{\text{DSM}}(\theta) = \tilde{\mathcal{J}}_{\text{ESM}}(\theta) + C$ for some C independent of θ . Consequently, the minimum point of $\tilde{\mathcal{J}}_{\text{DSM}}$ and that of $\tilde{\mathcal{J}}_{\text{ESM}}$ coincide.*

Proof. Let $\mathcal{J}_{\text{DSM}}(\theta) := \mathbb{E}_{X_0 \sim p_{data}(\cdot)} \mathbb{E}_{X_t | X_0} \left[|s_\theta(t, X_t) - \nabla \log p(t, X_t | X_0)|^2 \right]$. It suffices to prove that $\mathcal{J}_{\text{DSM}}(\theta) = \mathcal{J}_{\text{ESM}}(\theta) + C$ for some C independent of θ . Note that

$$\begin{aligned} \mathcal{J}_{\text{ESM}}(\theta) &= \mathbb{E}_{p(t, \cdot)} |s_\theta(t, X) - \nabla \log p(t, X)|^2 \\ &= \mathbb{E}_{p(t, \cdot)} \left[|s_\theta(t, X)|^2 - 2s_\theta(t, X)^\top \nabla \log p(t, X) + |\nabla \log p(t, X)|^2 \right]. \end{aligned}$$

For the inner product, we can rewrite it as:

$$\begin{aligned} \mathbb{E}_{p(t, \cdot)} [s_\theta(t, X)^\top \nabla \log p(t, X)] &= \int_x s_\theta(t, x)^\top \nabla p(t, x) dx \\ &= \int_x s_\theta(t, x)^\top \nabla \int_{x_0} p(0, x_0) p(t, x; x_0) dx_0 dx \\ &= \int_{x_0} \int_x s_\theta(t, x)^\top p(0, x_0) \nabla p(t, x; x_0) dx dx_0 \\ &= \int_{x_0} p(0, x_0) \int_x s_\theta(t, x)^\top p(t, x; x_0) \nabla \log p(t, \cdot; x_0) dx dx_0 \\ &= \mathbb{E}_{X_0 \sim p_{data}(\cdot)} \mathbb{E}_{X_t | X_0} [s_\theta(t, X_t)^\top \nabla \log p(t, X_t | X_0)]. \end{aligned}$$

Combining with $\mathbb{E}_{p(t, \cdot)} |s_\theta(t, X)|^2 = \mathbb{E}_{X_0 \sim p_{data}(\cdot)} \mathbb{E}_{X_t | X_0} |s_\theta(t, X)|^2$ concludes the proof. $\square$

The main takeaway of DSM is that the gradient of the log density at some corrupted point should ideally move towards the clean sample. As mentioned earlier, the conditional distribution of $(X_t | X_0)$ is required to be simple, e.g., Gaussian for all the examples in Section 3. Here we set:

$$(X_t | X_0) \sim \mathcal{N}(\mu_t(X_0), \sigma_t^2 I) \quad \text{for some } \mu_t(\cdot) \text{ and } \sigma_t > 0.$$

For instance, $\mu_t(x) = x$ and $\sigma_t = \sigma_{\min} \sqrt{\left(\frac{\sigma_{\max}}{\sigma_{\min}}\right)^{\frac{2t}{T}} - 1}$ for VE, and $\mu(t, x) = e^{-\frac{1}{2} \int_0^t \beta(s) ds} x$ and $\sigma_t = \sqrt{1 - e^{-\int_0^t \beta(s) ds}}$ for VP. In this case, we can compute explicitly the conditional score:

$$\nabla \log p(t, X_t | X_0) = \frac{\mu_t(X_0) - X_t}{\sigma_t^2}. \quad (33)$$

The direction $\frac{1}{\sigma_t^2}(X_t - \mu_t(X_0))$ clearly facilitates moving to the clean sample, and we want $s_\theta(t, x)$ to match the score (33) as best it can. The problem (32) reads as:

$$\begin{aligned} & \tilde{\mathcal{J}}_{\text{DSM}}(\theta) \\ &= \mathbb{E}_{t \sim \mathcal{U}(0, T)} \left\{ \lambda(t) \mathbb{E}_{X_0 \sim p_{\text{data}}(\cdot)} \mathbb{E}_{X_t | X_0} \left[\left| s_\theta(t, X_t) + \frac{X_t - \mu_t(X_0)}{\sigma_t^2} \right|^2 \right] \right\}. \end{aligned} \quad (34)$$

Alternatively, the problem (34) can be interpreted as the least squares of X_0 over X_t by *Tweedie's formula* [18]. In fact,

$$\mathbb{E}(\mu_t(X_0) | X_t) = X_t + \sigma_t^2 \nabla \log p(t, X_t). \quad (35)$$

So $\nabla \log p(t, x)$ minimizes the loss $\mathbb{E}_{X_0 \sim p_{\text{data}}(\cdot)} \mathbb{E}_{X_t | X_0} \left| h(t, X_t) + \frac{X_t - \mu_t(X_0)}{\sigma_t^2} \right|^2$ over all measurable function h . For function approximations, we replace h with $\{s_\theta(t, x)\}_\theta$. See also [87] for Tweedie's formula generalized to non-Gaussian diffusion models.

Empirically, it was observed [83] that a good candidate for the weight function $\lambda(t)$ is:

$$\lambda(t) \propto 1/\mathbb{E}|\nabla \log p(t, X_t | X_0)|^2 = \sigma_t^2. \quad (36)$$

The choice (36) is related to an evidence lower bound (ELBO) of the KL divergence between the generated distribution and the true distribution, see [33, 60, 80] for discussions. See also [43] for other choices of $\lambda(t)$. Injecting (36) into (34) yields the ultimate objective:

$$\begin{aligned} \tilde{\mathcal{J}}_{\text{DSM}}(\theta) &= \mathbb{E}_{t \sim \mathcal{U}(0, T)} \left\{ \sigma_t^2 \mathbb{E}_{X_0 \sim p_{\text{data}}(\cdot)} \mathbb{E}_{X_t | X_0} \left[\left| s_\theta(t, X_t) + \frac{X_t - \mu_t(X_0)}{\sigma_t^2} \right|^2 \right] \right\} \\ &= \mathbb{E}_{t \sim \mathcal{U}(0, T)} \left\{ \mathbb{E}_{X_0 \sim p_{\text{data}}(\cdot)} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} \left[|\sigma_t s_\theta(t, \mu_t(X_0) + \sigma_t \epsilon) + \epsilon|^2 \right] \right\}, \end{aligned} \quad (37)$$

where the second equality follows from a change of variables $\epsilon := \frac{X_t - \mu_t(X_0)}{\sigma_t}$.

Denosing score matching (37), which is essentially a least squares regression, is the most widely used approach for score matching. It often requires large batches to solve the stochastic optimization problem. Sliced score matching (31) is computationally efficient in high dimensions. However, its performance relies on the random projections, i.e., the choice of slices, which may incur high variance when the number of samples m is small.

5. ODE samplers and consistency model

The stochastic sampler (25) (and its discretization) may suffer from slow convergence and instability due to the randomness. Moreover, estimating the score function for each model is computationally expensive. In this section, we consider sampling Y_T by an equivalent deterministic scheme – the ODE sampler. Section 5.1 presents the probability flow ODE that leads to the ODE sampler of diffusion models. We also explain the reparametrization trick (mentioned at the end of Section 3), which avoids score matching repeatedly for different models. In Section 5.2, we discuss one prominent recent application of ODE sampling – the consistency model.

5.1. The probability flow ODE

The main idea is that starting from $Y'_0 \sim p(T, \cdot)$, and going through an ODE rather than an SDE will produce a sample $Y'_T \sim p_{\text{data}}(\cdot)$. Here we are only concerned with the sampled distribution at time T .

The equivalent ODE formulation is a consequence of the following result, known as the *probability flow ODE*.

Theorem 5.1 (Probability flow ODE). *Let $(X_t, 0 \leq t \leq T)$ be defined by (1), with $p(t, \cdot)$ the probability distribution of X_t . Let*

$$\tilde{f}(t, x) := f(t, x) - \frac{1}{2} \nabla \cdot (g(t, x)g(t, x)^\top) - \frac{1}{2} g(t, x)g(t, x)^\top \nabla \log p(t, x), \quad (38)$$

and let $(\tilde{X}_t, 0 \leq t \leq T)$ solve the ODE:

$$\frac{d\tilde{X}_t}{dt} = \tilde{f}(t, \tilde{X}_t), \quad \tilde{X}_0 \sim p_{\text{data}}(\cdot). \quad (39)$$

Then for each t , X_t and $\tilde{X}_t$ have the same distribution.

Proof. The idea is to match the Fokker-Planck equation in both settings. Denote by $\tilde{p}(t, x)$ the probability distribution of $\tilde{X}_t$ defined by (39). By the (first-order) Fokker-Planck equation, $\tilde{p}(t, x)$ solves

$$\frac{\partial \tilde{p}(t, x)}{\partial t} = -\nabla \cdot (\tilde{f}(t, x)\tilde{p}(t, x)), \quad \tilde{p}(0, x) = p_{\text{data}}(x). \quad (40)$$

On the other hand, $p(t, x)$ solves the (second order) Fokker-Planck equation:

$$\frac{\partial p(t, x)}{\partial t} = -\nabla \cdot (f(t, x)p(t, x)) + \frac{1}{2} \nabla^2 : (g(t, x)g(t, x)^\top p(t, x)), \quad p(0, x) = p_{\text{data}}(x). \quad (41)$$

where $:$ denotes the Frobenius inner product. By algebraic rearrangement, (41) simplifies to:

$$\frac{\partial p(t, x)}{\partial t} = -\nabla \cdot (\tilde{f}(t, x)p(t, x)), \quad p(0, x) = p_{\text{init}}(x). \quad (42)$$

Comparing (39) with (41) yields $\tilde{p}(t, \cdot) = p(t, \cdot)$. $\square$

We remark that the two processes X and $\tilde{X}$ only have the same marginal distributions, but they don't have the same distribution at the level of the process (nor the joint distributions).

Now let's apply the equivalent ODE formulation to the diffusion models. We specify to the practical case where $n = d$ and $\sigma(t, x) = \sigma(t)I$. The corresponding ODE is given by

$$\frac{dX_t}{dt} = f(t, X_t) - \frac{1}{2}g^2(t)\nabla \log p(t, X_t), \quad X_0 \sim p_{\text{data}}(\cdot), \quad (43)$$

and the backward ODE is then:

$$\frac{dY_t}{dt} = -f(T-t, Y_t) + \frac{1}{2}g^2(T-t)\nabla \log p(T-t, Y_t), \quad Y_0 \sim p_{\text{noise}}(\cdot). \quad (44)$$

With the score function $\nabla \log p(t, x)$ being replaced with the score matching function $s_\theta(t, x)$, the backward ODE becomes:

$$\frac{dY_t}{dt} = -f(T-t, Y_t) + \frac{1}{2}g^2(T-t)s_\theta(T-t, Y_t), \quad Y_0 \sim p_{\text{noise}}(\cdot), \quad (45)$$

which can be regarded as the continuum limit of *denoising diffusion implicit models* (DDIM) [77]. Among the examples in Section 3:

(b') The ODE sampler for VE is:

$$\frac{dY_t}{dt} = \frac{1}{2}g^2(T-t)s_\theta(T-t, Y_t), \quad Y_0 \sim p_{\text{noise}}(\cdot), \quad (46)$$

where g is defined by (11). An alternative parametrization is $g(t) = \sqrt{2t}$.

(c') The ODE sampler for VP is:

$$\frac{dY_t}{dt} = \frac{1}{2}\beta(T-t)(Y_t + s_\theta(T-t, Y_t)), \quad Y_0 \sim p_{\text{noise}}(\cdot). \quad (47)$$

where $\beta(t)$ is defined by (15).

The ODEs (46)–(47) can be easily solved by existing ODE solvers, see Section 6.3 for further discussions.

Now let's explain the reparametrization trick, which allows to sample from different models with the score function of VE. Let $p^{\text{VE}}(t, x)$ be the probability density of X_t , with $f(t, x) = 0$ and $g(t) = \sqrt{2t}$. Note that in all the above examples, $f(\cdot, \cdot)$ takes the form $f(t, x) = f(t)x$ with $f(t) \in \mathbb{R}$. In this case, by a change of variables [41], the diffusion model can be reparametrized by

$$s(t) := \exp\left(\int_0^t f(r)dr\right) \quad \text{and} \quad \ell(t) := \sqrt{\int_0^t \frac{g(r)^2}{s(r)^2}dr}, \quad (48)$$

so that the ODE (43) becomes:

$$\frac{dX_t}{dt} = \frac{s'(t)}{s(t)}X_t - s(t)^2\ell'(t)\ell(t)\nabla \log p^{\text{VE}}\left(\ell(t), \frac{X_t}{s(t)}\right), \quad X_0 \sim p_{\text{data}}(\cdot). \quad (49)$$

That is, a diffusion model with $f(t, x) = f(t)x$ can be represented in terms of VE by a suitable space-time change. For instance, $s(t) = 1$ and $\ell(t) = t$ for VE with $f(t) = 0$ and $g(t) = \sqrt{2t}$, and $s(t) = e^{-\frac{1}{2} \int_0^t \beta(r) dr}$ and $\ell(t) = \sqrt{e^{\int_0^t \beta(r) dr} - 1}$ for VP with $f(t) = -\frac{1}{2}\beta(t)$ and $g(t) = \sqrt{\beta(t)}$.

Using the score matching techniques in Section 4, we can approximate the score $\nabla \log p^{\text{VE}}(t, x)$ by a class of functions $\{s_\theta^{\text{VE}}(t, x)\}_\theta$. The ODE sampler is then:

$$\begin{aligned} \frac{dY_t}{dt} &= -\frac{s'(T-t)}{s(T-t)}Y_t + s(T-t)^2\ell'(T-t)\ell(T-t)s_\theta^{\text{VE}}\left(\ell(T-t), \frac{Y_t}{s(T-t)}\right), \\ Y_0 &\sim p_{\text{noise}}(\cdot). \end{aligned} \quad (50)$$

In particular, the ODE sampler for VP is:

$$\begin{aligned} \frac{dY_t}{dt} &= \frac{1}{2}\beta(T-t)\left(Y_t + s_\theta^{\text{VE}}\left(\sqrt{e^{\int_0^{T-t}\beta(r)dr} - 1}, Y_t e^{\frac{1}{2}\int_0^{T-t}\beta(r)dr}\right)\right), \\ Y_0 &\sim p_{\text{noise}}(\cdot). \end{aligned}$$

5.2. Consistency model

Here we present an application of the probability flow ODE – the consistency model [79] (see [42, 78] for a more advanced version, and [61] for its latest application.) The advantage of the consistency modeling is that it only performs a *one-step* sampling – this is in contrast with the SDE or the ODE sampling, where multiple steps are required in discretization.

The main takeaway of the consistency model is to approximate the ODE flow. Recall the ODE sampler from (44). The ODE theory implies that there is a function (the ODE flow) $F : \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that

$$Y_t = F(t, s, Y_s), \quad \text{for } 0 \leq s < t \leq T. \quad (51)$$

Thus, $Y_T = F(T, t, Y_t)$ for any t . The consistency model proposes to learn $F(T, \cdot, \cdot)$ directly. To be more precise, the function of form $F_\theta(T-t, y)$ is used to approximate the flow $F(T, t, y)$. (Note that if the ODE (44) is autonomous, e.g. the OU process, the flow $F(t, s, y)$ is indeed of form $F(t-s, y)$.) With the learned flow F_θ , the model outputs the sample $F_\theta(T, Y_0)$, $Y_0 \sim p_{\text{noise}}(\cdot)$ in just one step.

Now the question is how to learn the ODE flow $F(T, t, y) \approx F_\theta(T-t, y)$. Two methods were proposed based on the group property (51).

(a) *Consistency distillation* (CD): the approach builds on top of a pretrained score function $s_\theta(t, x)$. Take $t \sim \mathcal{U}(0, T)$. First sample $Y_+ \stackrel{d}{=} X_t$ of the forward process (VE with $g(t) = \sqrt{2t}$ is used, so $X_t \sim \mathcal{N}(x, t^2 I)$ with $x \sim p_{\text{data}}(\cdot)$). Then use the ODE sampler (44) to set:

$$Y_- = Y_+ + \left(-f(t, Y_+) + \frac{1}{2}g^2(t)s_\theta(t, Y_+)\right)\delta, \quad (52)$$

for some small δ . Roughly speaking, (Y_+, Y_-) are on the ODE flow F at the times $T - t$ and $T - t + \delta$. Inspired by the fact that

$$\underbrace{F(T, T - t, Y_{T-t})}_{\approx F_\theta(t, Y_+)} = Y_T = \underbrace{F(T, T - t + \delta, Y_{T-t+\delta})}_{\approx F_\theta(t-\delta, Y_-)},$$

the training objective is:

$$\mathbb{E}_{t \in \mathcal{U}(0, T)} \mathbb{E}[\lambda(t) |F_\theta(t, Y_+) - F_\theta(t - \delta, Y_-)|^2], \quad (53)$$

where $\lambda(t)$ is a weight function.

(b) *Consistency training* (CT): the approach is similar to CD, except that $Y_- \stackrel{d}{=} X_{t-\delta}$ is taken *independently* of $Y_+ \stackrel{d}{=} X_t$. (In the VE setting with $g(t) = \sqrt{2t}$, $Y_+ \sim \mathcal{N}(x, t^2 I)$ and $Y_- \sim \mathcal{N}(x, (t - \delta)^2 I)$ independently.) Thus, CT does not require a pretrained score function. Note that

$$Y_+ \stackrel{d}{=} Y_{T-t} \quad \text{and} \quad Y_- \stackrel{d}{=} Y_{T-t+\delta} \quad \text{marginally.}$$

But in contrast with CD, (Y_+, Y_-) are not on the ODE flow path, i.e., a wrong coupling.

Now we consider a simple scenario, where $g(t) = \sqrt{2t}$ and $p_{\text{data}}(\cdot) = \delta_0$ (delta mass at 0). It is easy to compute that

$$\begin{aligned} (Y_+^{CT}, Y_-^{CT}) &\stackrel{d}{=} \mathcal{N}\left(0, \begin{pmatrix} t^2 I & 0 \\ 0 & (t - \delta)^2 I \end{pmatrix}\right), \\ (Y_+^{CD}, Y_-^{CD}) &\stackrel{d}{=} \mathcal{N}\left(0, \begin{pmatrix} t^2 I & t(t - \delta)I \\ t(t - \delta)I & (t - \delta)^2 I \end{pmatrix}\right). \end{aligned}$$

As a result,

$$\begin{aligned} W_2^2((Y_+^{CD}, Y_-^{CD}), (Y_+^{CT}, Y_-^{CT})) &= 2d \left(t^2 + (t - \delta)^2 - \sqrt{t^4 + (t - \delta)^4} \right) \\ &\approx 2(2 - \sqrt{2})t^2 d. \end{aligned} \quad (54)$$

So the distance between (Y_+^{CD}, Y_-^{CD}) and (Y_+^{CT}, Y_-^{CT}) is far from being negligible. By taking $t \sim \mathcal{U}(0, 1)$, the expected W_2 distance is of order $\sqrt{\left(1 - \frac{\sqrt{2}}{2}\right)} d$. This explains why CD outperforms CT empirically [79]. A remedy to CT has recently been proposed in [78].

Also note that the objective function (53) can be lifted to continuous time via partial differential relation of the ODE flow $Y_T = F(T, T - t, Y_{T-t}) \approx F_\theta(t, Y_{T-t})$:

$$\mathbb{E}_{t \in \mathcal{U}(0, T)} \mathbb{E} \left[\lambda(t) \left| \partial_t F_\theta(t, Y_+) + \nabla F_\theta(t, Y_+) \left(f(t, Y_+) - \frac{1}{2} g^2(t) s_\theta(t, Y_+) \right) \right|^2 \right], \quad (55)$$

or

$$\mathbb{E}_{t \in \mathcal{U}(0,T)} \mathbb{E} \left[\lambda(t) F_\theta(t, Y_+) \cdot \left(\partial_t F_\theta(t, Y_+) + \nabla F_\theta(t, Y_+) \frac{Y_+ - x}{t} \right) \right], \quad (56)$$

where $Y_+ \stackrel{d}{=} X_t$ and $x \sim p_{\text{data}}(\cdot)$. The functions (55) and (56) are the continuum limits of CD and CT respectively after suitable scaling.

6. Convergence results

In this section, we consider the convergence of the stochastic samplers, i.e., the backward processes, under a “blackbox” assumption on score matching:

Assumption 6.1. *There exists $\varepsilon > 0$ such that $\mathbb{E}_{p(t,\cdot)} |s_\theta(t, X) - \nabla \log p(t, X)|^2 \leq \varepsilon^2$, $t \in [0, T]$ for some $s_\theta(\cdot, \cdot)$.*

For instance, the score matching function $s_\theta(\cdot, \cdot)$ can be the output of any algorithm introduced in Section 4. This condition postulates an L^2 -bound on score matching. As we will see in Section 7, there has been a body of work on the accuracy of score matching. So these assumptions, especially Assumption 6.1 can be replaced with those score approximation bounds.

The quality of the diffusion model relies on how $(f(\cdot, \cdot), g(\cdot))$ is chosen. Since the SDEs are inherently in continuous time, the convergence of the diffusion model as a continuous process yields another criterion on the design of the model. Ideally, we want to show that the stochastic sampler Y_T defined by (24), as well as its discretization are close in distribution to the target distribution $p_{\text{data}}(\cdot)$. Here we focus on two metrics: the total variation distance and the Wasserstein-2 distance. While most existing convergence results were established in the total variation distance (or KL divergence), the convergence in the Wasserstein metric is important because it has shown to align with human judgment on image similarity [5], and the standard evaluation metric – Fréchet inception distance (FID) is also based on the Wasserstein distance.

For simplicity, we assume (as in Section 3) that $n = d$ and $g(t, x) = g(t)I$. In Sections 6.1–6.2, we study the convergence of the stochastic samplers in continuous time in the two metrics respectively. The problem of discretization will be discussed in Section 6.3, with additional references given.

6.1. Stochastic sampler – total variation bound

The goal is to bound $d_{TV}(\mathcal{L}(Y_T), p_{\text{data}}(\cdot))$, with Y defined by (25). [15] first provided a bound of $d_{TV}(\mathcal{L}(Y_T), p_{\text{data}}(\cdot))$ for the OU process (7), assuming an L^∞ -bound on score matching³. The bound is of form:

$$d_{TV}(\mathcal{L}(Y_T), p_{\text{data}}(\cdot)) \leq \underbrace{d_{TV}(p(T, \cdot), p_{\text{noise}}(\cdot))}_{\leq C \exp(-C'T)} + D\varepsilon \exp(D'T), \quad (57)$$

³There exists $\varepsilon > 0$ such that $|s_\theta(t, \cdot) - \nabla \log p(t, \cdot)|_\infty \leq \varepsilon$, $t \in [0, T]$ for some $s_\theta(\cdot, \cdot)$.

for some $C, C', D, D' > 0$ depending on θ, μ and $p_{\text{data}}(\cdot)$. The disadvantage of this result is that the score matching error grows exponentially in time T . A recent breakthrough [11] improved this bound to be polynomial in T (and d). The result is stated for the (discretized) OU process, but is easily extended to the general case.

Theorem 6.2 (Total variation bound). *[11] Let Assumption 6.1 hold, and assume that $g(t)$ is bounded away from zero. Then*

$$d_{TV}(\mathcal{L}(Y_T), p_{\text{data}}(\cdot)) \leq \underbrace{d_{TV}(p(T, \cdot), p_{\text{noise}}(\cdot))}_{\text{noise approx. error}} + \underbrace{\varepsilon\sqrt{T/2}}_{\text{score matching error}}. \quad (58)$$

Proof. We sketch the proof, which is to decompose $d_{TV}(\mathcal{L}(Y_T), p_{\text{data}}(\cdot))$ into the noise approximation error and the score matching error. Denote by

- Q_T the distribution of the backward process (25), i.e., with the matched score $s_\theta(t, x)$ and $Y_0 \sim p_{\text{noise}}(\cdot)$;
- Q'_T the distribution of the backward process, with the matched score $s_\theta(t, x)$ and $Y_0 \sim p(T, \cdot)$;
- Q''_T the distribution of the backward process, with the true score $\nabla \log p(t, x)$ and $Y_0 \sim p(T, \cdot)$.

By the data processing inequality, we have:

$$\begin{aligned} d_{TV}(\mathcal{L}(Y_T), p_{\text{data}}(\cdot)) &\leq d_{TV}(Q_T, Q'_T) + d_{TV}(Q'_T, Q''_T) \\ &\leq d_{TV}(p(T, \cdot), p_{\text{noise}}(\cdot)) + d_{TV}(Q'_T, Q''_T). \end{aligned} \quad (59)$$

The key to bound the term $d_{TV}(Q'_T, Q''_T)$ is the identity:

$$\text{KL}(Q''_T, Q'_T) = \mathbb{E}_{Q''_T} \left(\log \frac{dQ''_T}{dQ'_T} \right) = \mathbb{E}_{Q''_T} \int_0^T |s_\theta(T-t, Y_t) - \nabla \log p(T-t, Y_t)|^2 dt, \quad (60)$$

where the second equality follows from the Girsanov theorem. Further by Assumption 6.1, we get by Pinsker's inequality:

$$d_{TV}(Q'_T, Q''_T) \leq \sqrt{\frac{1}{2} \text{KL}(Q''_T, Q'_T)} \leq \varepsilon \sqrt{T/2}. \quad (61)$$

Combining (59) and (61) yields (58). $\square$

Now let's consider an important special case – the VP SDE that also includes the OU process.

Corollary 6.3 (Total variation for VP). *Let $(Y_t, 0 \leq t \leq T)$ be specified by (14) (the backward process of VP). Let Assumption 6.1 hold, and assume that $\beta(t)$ is bounded away from zero, and $\mathbb{E}_{p_{\text{data}}(\cdot)} |x|^2 < \infty$. Then for T sufficiently large,*

$$d_{TV}(\mathcal{L}(Y_T), p_{\text{data}}(\cdot)) \leq e^{-\frac{1}{2} \int_0^T \beta(s) ds} \sqrt{\mathbb{E}_{p_{\text{data}}(\cdot)} |x|^2 / 2} + \varepsilon \sqrt{T/2}. \quad (62)$$

Proof. Recall from (16) that for VP, the conditional distribution of $(X_t | X_0 = x)$ is $\mathcal{N}\left(e^{-\frac{1}{2} \int_0^t \beta(s) ds} x, (1 - e^{-\int_0^t \beta(s) ds}) I\right)$. By the convexity of KL divergence, we have:

$$\begin{aligned} & \text{KL}(p(T, \cdot), p_{\text{noise}}(\cdot)) \\ & \leq \mathbb{E}_{p_{\text{data}}(\cdot)} \left[\underbrace{\text{KL}\left(\mathcal{N}\left(e^{-\frac{1}{2} \int_0^T \beta(s) ds} x, (1 - e^{-\int_0^T \beta(s) ds}) I\right), \mathcal{N}(0, I)\right)}_{(a)} \right]. \end{aligned} \quad (63)$$

Moreover,

$$\begin{aligned} (a) &= e^{-\int_0^T \beta(s) ds} |x|^2 - d\left(e^{-\int_0^T \beta(s) ds} + \log(1 - e^{-\int_0^T \beta(s) ds})\right) \\ &\leq e^{-\int_0^T \beta(s) ds} |x|^2, \quad \text{as } T \rightarrow \infty, \end{aligned} \quad (64)$$

since $\beta(t)$ is bounded away from zero, and hence $\int_0^T \beta(s) ds \rightarrow \infty$ as $T \rightarrow \infty$. Combining (63), (64) with (58) yields (62). $\square$

Similar polynomial bounds were also obtained by [47, 48] for the VP SDE. There χ^2 -divergence (and an identity analogous to (60)) is used to bound $d_{TV}(\mathcal{L}(Y_T), p_{\text{data}}(\cdot))$.

6.2. Stochastic sampler – Wasserstein bound

We make the following additional assumptions.

Assumption 6.4. *The following conditions hold:*

1. *There exists $r_f : [0, T] \rightarrow \mathbb{R}$ such that $(x - x') \cdot (f(t, x) - f(t, x')) \geq r_f(t) |x - x'|^2$ for all t, x, x' .*
2. *There exists $L > 0$ such that $|s_\theta(t, x) - s_\theta(t, x')| \leq L |x - x'|$ for all t, x, x' .*

The condition (1) assumes the growth of $f(t, \cdot)$, and (2) assumes the Lipschitz property of the score matching function (as considered in [46]).

Theorem 6.5 (Wasserstein bound). *[89] Let Assumptions 6.1, 6.4 hold. For $h > 0$ a hyperparameter, define*

$$u(t) := \int_{T-t}^T (-2r_f(s) + (2L + 2h)g^2(s)) ds. \quad (65)$$

Then we have

$$W_2(p_{\text{data}}(\cdot), \mathcal{L}(Y_T)) \leq \sqrt{\underbrace{W_2^2(p(T, \cdot), p_{\text{noise}}(\cdot)) e^{u(T)}}_{\text{noise approx. error}} + \underbrace{\frac{\varepsilon^2}{2h} \int_0^T g^2(t) e^{u(T)-u(T-t)} dt}_{\text{score matching error}}}. \quad (66)$$

Proof. The idea relies on coupling. Consider the coupled SDEs:

$$\begin{cases} dU_t = (-f(T-t, U_t) + g^2(T-t)\nabla \log p(T-t, U_t)) dt + g(T-t)dB_t, \\ dV_t = (-f(T-t, V_t) + g^2(T-t)s_\theta(T-t, V_t)) dt + g(T-t)dB_t, \end{cases}$$

where (U_0, V_0) are coupled to achieve $W_2(p(T, \cdot), p_{\text{noise}}(\cdot))$. Note that

$$W_2^2(p_{\text{data}}(\cdot), \mathcal{L}(Y_T)) \leq \mathbb{E}|U_T - V_T|^2,$$

so the goal is to bound $\mathbb{E}|U_T - V_T|^2$. By Itô's formula, we get

$$\begin{aligned} d|U_t - V_t|^2 &= 2(U_t - V_t) \cdot (-f(T-t, U_t) + g^2(T-t)\nabla \log p(T-t, U_t) \\ &\quad + f(T-t, V_t) - g^2(T-t)s_\theta(T-t, V_t))dt, \end{aligned}$$

which implies that

$$\begin{aligned} \frac{d\mathbb{E}|U_t - V_t|^2}{dt} &= -2\mathbb{E}((U_t - V_t) \cdot (f(T-t, U_t) - f(T-t, V_t))) \\ &\quad + 2\mathbb{E}((U_t - V_t) \cdot g^2(T-t)(\nabla \log p(T-t, U_t) - s_\theta(T-t, V_t))). \end{aligned} \tag{67}$$

By Assumption 6.4 (1), we get (a) $\geq r_f(T-t) \mathbb{E}|U_t - V_t|^2$. Moreover,

$$\begin{aligned} (b) &= g^2(T-t) \left(\mathbb{E}((U_t - V_t) \cdot (\nabla \log p(T-t, U_t) - s_\theta(T-t, U_t))) \right. \\ &\quad \left. + \mathbb{E}((U_t - V_t) \cdot (s_\theta(T-t, U_t) - s_\theta(T-t, V_t))) \right) \\ &\leq g^2(T-t) \left(h\mathbb{E}|U_t - V_t|^2 + \frac{1}{4h}\varepsilon^2 + L\mathbb{E}|U_t - V_t|^2 \right), \end{aligned} \tag{68}$$

where we use Assumptions 6.1 and 6.4 (2) in the last inequality. Therefore,

$$\frac{d\mathbb{E}|U_t - V_t|^2}{dt} \leq (-2r_f(T-t) + (2h + 2L)g^2(T-t)) \mathbb{E}|U_t - V_t|^2 + \frac{\varepsilon^2}{2h}g^2(T-t).$$

Applying Grönwall's inequality, we have $\mathbb{E}|U_T - V_T|^2 \leq e^{u(T)}\mathbb{E}|U_0 - V_0|^2 + \frac{\varepsilon^2}{2h} \int_0^T \sigma^2(T-t)e^{u(T)-u(t)}dt$, which yields (66). $\square$

Theorem 6.5 assumes that the score matching function is Lipschitz, which may be too restrictive, say for a neural net. A more natural assumption is that there exists $L > 0$ such that

$$|\nabla \log p(t, x) - \nabla \log p(t, x')| \leq L|x - x'| \text{ for all } t, x, x',$$

i.e., the score function (rather than the score matching function) is Lipschitz. In this case, the result still holds if an L^∞ bound on score matching is assumed.

[24] obtained a similar result by an L^2 -bound on score matching but directly assuming that $\mathbb{E}|s_\theta(T-t, V_t) - \nabla \log p(T-t, V_t)|^2 \leq \varepsilon^2$ for all t .

Theorem 6.5 does not require a specific structure of $f(t, x)$. The bound (66) is loose in the general case, e.g., $r_f(t)$ may be negative, and hence the bound is exponential in T . Nevertheless, the estimates in the proof can be refined in some special cases. Here we examine the VP model.

The following result provides a bound for VP, assuming that $p_{\text{data}}(\cdot)$ is strongly log-concave. Recall that a smooth function $h : \mathbb{R}^d \rightarrow \mathbb{R}$ is κ -strongly log-concave if

$$(\nabla \log h(x) - \nabla \log h(y)) \cdot (x - y) \leq -\kappa|x - y|^2. \quad (69)$$

In comparison with Corollary 6.3, the score matching error does not grow in T .

Theorem 6.6 (Wasserstein bound for VP). *Let $(Y_t, 0 \leq t \leq T)$ be defined by (14) (the backward process of VP). Assume that $p_{\text{data}}(\cdot)$ is κ -strongly log-concave with $\kappa > \frac{1}{2}$, and $\mathbb{E}_{p_{\text{data}}(\cdot)}|x|^2 < \infty$. For $h < \min(\kappa, 1) - \frac{1}{2}$, we have:*

$$W_2^2(p_{\text{data}}(\cdot), \mathcal{L}(Y_T)) \leq e^{-\{\beta_{\min} + \beta_{\max}(2\min(\kappa, 1) - 1 - 2h)\}T} \left(\mathbb{E}_{p_{\text{data}}(\cdot)}|x|^2 + o(e^{-\beta_{\min}T})d \right) + \varepsilon^2 \frac{\beta_{\max}^2}{2h(2\min(\kappa, 1) - 1 - 2h)}. \quad (70)$$

Proof. Recall that $r_f(t) = -\frac{1}{2}\beta(t)$, $g(t) = \sqrt{\beta(t)}$ and $p_{\text{noise}}(\cdot) \sim \mathcal{N}(0, I)$. The key improvement relies on the fact [24, Proposition 10]: If $p_{\text{data}}(\cdot)$ is κ -strongly log-concave, then $p(T-t, \cdot)$ is

$$\kappa \left(e^{-\int_0^{T-t} \beta(s)ds} + \kappa \int_0^{T-t} e^{-\int_s^{T-t} \beta(v)dv} \beta(s)ds \right)^{-1} - \text{strongly log-concave.}$$

(The convolution preserves strong log-concavity.) Thus, the term $\mathbb{E}((U_t - V_t) \cdot (\nabla \log p(T-t, U_t) - \nabla \log p(T-t, V_t)))$ in (68) is bounded from above by

$$-\frac{\kappa}{e^{-\int_0^{T-t} \beta(s)ds} + \kappa \int_0^{T-t} e^{-\int_s^{T-t} \beta(v)dv} \beta(s)ds} \mathbb{E}|U_t - V_t|^2.$$

instead of $L \mathbb{E}|U_t - V_t|^2$. Consequently, we obtain the bound (66) by replacing $u(t)$ with

$$u_{\text{VP}}(t) := \int_{T-t}^T \beta(s) \left(1 + 2h - \frac{2\kappa}{e^{-\int_0^s \beta(v)dv} + \kappa \int_0^s e^{-\int_v^s \beta(u)du} \beta(v)dv} \right) ds. \quad (71)$$

It is easy to see that $u_{\text{VP}}(T) \leq \beta_{\max}(1 + 2h - 2\min(\kappa, 1))T$, and $u_{\text{VP}}(T-t) \leq \beta_{\max}(1 + 2h - 2\min(\kappa, 1))t$. Moreover,

$$\begin{aligned} W_2^2(p(T, \cdot), p_{\text{noise}}(\cdot)) &\leq e^{-\int_0^T \beta(s)ds} \mathbb{E}_{p_{\text{data}}(\cdot)}|x|^2 + \frac{d}{4} e^{-2\int_0^T \beta(s)ds} \\ &\leq e^{-\beta_{\min}T} \mathbb{E}_{p_{\text{data}}(\cdot)}|x|^2 + \frac{d}{4} e^{-2\beta_{\min}T}. \end{aligned}$$

For $\kappa > \frac{1}{2}$, we take $h < \min(\kappa, 1) - \frac{1}{2}$, so $1 + 2h - 2\min(\kappa, 1) < 0$. Combining (66) with the above estimates yields (70). $\square$

See [89] for a Wasserstein bound for CVP. In comparison with VP, CVP (and more general CDPM) is designed to be robust to the score matching error and discretization (see Section 6.3), at the cost of noise approximation error. Practically, T should not set to be too large for CVP to control the noise approximation bias.

6.3. Discretization

In order to implement the diffusion model, we need to discretize the corresponding SDE. For the SDE (25), there are several ways of discretization. Fix $0 = t_0 < t_1 \cdots < t_N = T$ to be the time discretization. Initialize $\hat{X}_0 \sim p_{\text{noise}}(\cdot)$, and set

$$\begin{aligned} \hat{X}_k &= \hat{X}_{k-1} + (-f(T - t_k, \hat{X}_{k-1}) + g^2(T - t_{k-1})s_\theta(T - t_{k-1}, \hat{X}_{k-1}))(t_k - t_{k-1}) \\ &\quad + \int_{t_{k-1}}^{t_k} g(T - t)dB_t, \quad \text{for } k = 1, \dots, N, \end{aligned} \quad (72)$$

where $\int_{t_{k-1}}^{t_k} g(T - t)dB_t$ is sampled as $\mathcal{N}(0, \int_{t_{k-1}}^{t_k} g^2(T - t)dt)$, or the Euler-Maruyama scheme:

$$\begin{aligned} \hat{X}_k &= \hat{X}_{k-1} + (-f(T - t_k, \hat{X}_{k-1}) + g^2(T - t_{k-1})s_\theta(T - t_{k-1}, \hat{X}_{k-1}))(t_k - t_{k-1}) \\ &\quad + g(T - t_{k-1})(B_{t_k} - B_{t_{k-1}}), \quad \text{for } k = 1, \dots, N. \end{aligned} \quad (73)$$

In practice, the predictor-corrector (PC) sampler [83] is frequently used, where the scheme (72) or (73) serves as the predictor, followed by a score-based corrector. More precisely, for each k , let $\hat{X}_k^{(0)}$ be the output of the predictor (72) or (73), and fix some $\epsilon_k > 0$. The corrector $\hat{X}_k = \hat{X}_k^{(M)}$ is given by

$$\hat{X}_k^{(\ell+1)} = \hat{X}_k^{(\ell)} + \epsilon_k s_\theta(T - t_k, \hat{X}_k^{(\ell)}) + \sqrt{2}(B'_{(\ell+1)\epsilon_k} - B'_{\ell\epsilon_k}), \quad \text{for } \ell = 0, \dots, M-1, \quad (74)$$

where M is the number of steps in the corrector step. The hyperparameters ϵ_k are taken to be decreasing in k . Essentially, (74) is the discrete scheme of the Langevin equation $dX_t = \nabla \log p(T - t_k, X_t) + \sqrt{2}B'_t$ that converges to $p(T - t_k, \cdot)$, but with the score function replaced with the estimated $s_\theta(T - t_k, \cdot)$. Since the scheme (72) or (73) is the predictor of the PC sampler, it is also called the predictor-only sampler.

Again as seen in Section 3, most popular diffusion models have a common structure: $f(t, x) = f(t)x$, with $f(t) \in \mathbb{R}$. In this case, the exponential integrator (EI) [99] can be applied to solve the SDE (25), or the ODE (45). Letting $\Psi(s, t) := e^{-\int_s^t f(T-u)du}$, we have for $0 \leq s < t \leq T$,

$$Y_t = \Psi(s, t)Y_s + \int_s^t \Psi(u, t)g^2(T - u)s_\theta(T - u, Y_u)du + \int_s^t \Psi(u, t)g(T - u)dB_u, \quad (25')$$

or

$$Y_t = \Psi(s, t)Y_s + \frac{1}{2} \int_s^t \Psi(u, t)g^2(T - u)s_\theta(T - u, Y_u)du. \quad (45')$$

So the discretization is:

$$\begin{aligned} \hat{X}_k = \Psi(t_{k-1}, t_k)\hat{X}_{k-1} &+ \left(\int_{t_{k-1}}^{t_k} \Psi(t_{k-1}, u)g^2(T - u)du \right) s_\theta(T - t_{k-1}, \hat{X}_{k-1}) \\ &+ \int_{t_{k-1}}^{t_k} \Psi(t_{k-1}, u)g(T - u)dB_u, \quad \text{for } k = 1, \dots, N. \end{aligned} \quad (75)$$

or

$$\begin{aligned} \hat{X}_k = \Psi(t_{k-1}, t_k)\hat{X}_{k-1} &+ \frac{1}{2} \left(\int_{t_{k-1}}^{t_k} \Psi(t_{k-1}, u)g^2(T - u)du \right) s_\theta(T - t_{k-1}, \hat{X}_{k-1}), \\ &\text{for } k = 1, \dots, N. \end{aligned} \quad (76)$$

Note that the EI solvers (75)–(76) for VP correspond to DDPM and DDIM, respectively. See also [97, 100] for higher order EI solvers.

Classical SDE theory [44] indicates that the discretization error is of order $C(T)\delta$, with $C(T)$ exponential in T . A natural question is whether there are better bounds for the SDEs arising from the diffusion models in Section 3. The general way to control the discretization error is to prove a one-step estimate, and then unfold the recursion. The analysis is mostly tedious, and we do not plan to expand the details here. The table below summarizes known results on the discretization error of various diffusion models.

TABLE 1
Discretization errors.

Ref.	Metric	Model	Sampling	Step size	Discretization error
[11]	TV	OU	Euler	δ	$\sqrt{Td}\sqrt{\delta}$
[47, 48]	TV	OU	PC	varying	$\mathcal{O}_T(1)\sqrt{d}\sqrt{\delta}$
[97]	TV	OU	EI	δ	$\sqrt{Td^7}\sqrt{\delta^3}$
[24]	W_2	VE	Euler	δ	$e^{\mathcal{O}(T)}\sqrt{d}\delta$
	W_2	VP	Euler	δ	$e^{\mathcal{O}(T^3)}\sqrt{d}\delta$
[89]	W_2	CDPM	Euler	δ	$\mathcal{O}_T(1)\sqrt{d}\sqrt{\delta}$

There has been work on the convergence of diffusion models in a compact domain, or satisfying the manifold hypothesis [14, 67]. In this case, the Wasserstein distance is simply bounded by the total variation distance. [11, 47, 48] also considered bounded target distributions, which should not be confused with diffusion models in a compact domain because the SDEs can live in an unbounded space. An early stopping argument is used there to guarantee the convergence of diffusion models in the W_2 metric. See also [2] for finer results on the convergence of the OU process, and [50, 51, 53] that worked directly on the convergence of (discrete) DDPM.

The theory of the ODE sampling of diffusion models is, however, still sparse. [10, 34] studied the total variation convergence of the ODE sampler for the OU process, and [26] established Wasserstein bounds for the ODE sampler of VE and VP. See also [3, 12, 51, 53, 55] for closely related results on the deterministic sampler of diffusion models.

Recently, there has been a surge of interest in understanding the convergence of diffusion models, where the target distribution is concentrated on low-dimensional manifolds in a higher-dimensional space. This phenomenon is called “adapting to unknown low-dimensional structures” [54]. Typically, the dependence of the discretization error in d is replaced with some intrinsic dimension $k \ll d$ (plus a polynomial $\log d$ correction). See also [35, 56, 68, 85] for further development in this direction.

7. Further results on score matching

In the previous sections, we have seen that the success of diffusion models relies largely on the quality of score matching. This section reviews further results on score matching in the context of diffusion models. Section 7.1 considers statistical efficiency of score matching, and Section 7.2 presents approximations by neural nets. In Section 7.3, we introduce the idea of diffusion model alignment by reinforcement learning (RL).

7.1. Statistical efficiency of score matching

In this subsection, we focus on the statistical aspect of the score matching problem (27). Assume that

$$s_\theta(t, x) = \nabla \log p_\theta(t, x), \quad \text{for all } t, x, \quad (77)$$

where $p_\theta(t, \cdot)$ is a family of probability distributions parametrized by θ . In other words, $s_\theta(\cdot, \cdot)$ is the score function of a parametric family of probability distributions instead of an arbitrary neural net. This assumption is restrictive (and may seem to not be realistic), but it serves as a first step to understand the statistical efficiency of score matching in the context of diffusion models.

Under the assumption (77), the score matching problem (27) becomes

$$\min_{\theta} \mathcal{J}_{\text{ESM}}(\theta) := \mathbb{E}_{p(t, \cdot)} |\nabla \log p_\theta(t, x) - \nabla \log p(t, X)|^2, \quad (78)$$

and the corresponding implicit score matching problem is:

$$\begin{aligned} \min_{\theta} \mathcal{J}_{\text{ISM}}(\theta) &:= \mathbb{E}_{p(t, \cdot)} [|\nabla \log p_\theta(t, X)|^2 + 2 \nabla \cdot \nabla \log p_\theta(t, X)] \\ &= \mathbb{E}_{p(t, \cdot)} [|\nabla \log p_\theta(t, X)|^2 + 2 \text{Tr}(\nabla^2 \log p_\theta(t, X))] . \end{aligned} \quad (79)$$

Let $\hat{\theta}_n$ be minimum point of the empirical version of (78) or (79) from n samples $X_1, \dots, X_n \sim p(t, \cdot)$:

$$\hat{\theta}_n := \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n (|\nabla \log p_\theta(t, X_i)|^2 + 2 \text{Tr}(\nabla^2 \log p_\theta(t, X_i))) . \quad (80)$$

The question of interest is whether the estimated $p_{\hat{\theta}_n}(t, \cdot)$ approximates well the probability distribution $p(t, \cdot)$. The answer is hinted in the following theorem.

Theorem 7.1 (Consistency and asymptotic normality). *[37, 45] Fix $t \in [0, T]$.*

1. *Assume that $p(t, \cdot)$ has full support, and $p(t, \cdot) = p_{\theta_*}(t, \cdot)$ for some θ_* and $p(t, \cdot) \neq p_{\theta}(t, \cdot)$ for any $\theta \neq \theta_*$. Then $\mathcal{J}_{ESM}(\theta) = 0$ if and only if $\theta = \theta_*$.*
2. *Assume further that $p_{\theta}(t, \cdot)$ belongs to the exponential family, i.e. $p_{\theta}(t, \cdot) \propto \exp(\theta \cdot F(\cdot))$ for a suitably nice F . Then*

$$\sqrt{n}(\hat{\theta}_n - \theta_*) \rightarrow \mathcal{N}(0, \Gamma_{SM}), \quad (81)$$

with

$$\Gamma_{SM} := \mathbb{E}_{p_{\theta_*}(t, \cdot)} \left[(\nabla F(X) \nabla F(X)^{\top})^{-1} \right. \\ \left. \Sigma_{\nabla F(X) \nabla F(X)^{\top} \theta_* + \Delta F(X)} \mathbb{E}(\nabla F(X) \nabla F(X)^{\top})^{-1} \right],$$

where $\Sigma_{\bullet}$ is the Fisher information matrix, and the Laplacian $\Delta F(X)$ applies coordinate-wise to $F(X)$.

Proof. (1) If $\mathcal{J}_{ESM}(\theta) = 0$, then $\nabla \log p_{\theta}(t, \cdot) = \nabla \log p(t, \cdot)$. So $p_{\theta}(t, \cdot) = p(t, \cdot) + C$, and $C = 0$ because $p(t, \cdot)$ and $p_{\theta}(t, \cdot)$ are probability distributions. By the hypothesis, we have $\theta = \theta_*$.

(2) Recall that $p(t, \cdot) = p_{\theta_*}(t, \cdot)$. The key to the analysis relies on the fact that

$$\theta_* = -\mathbb{E}(\nabla F(X) \nabla F(X)^{\top})^{-1} \mathbb{E} \Delta F(X). \quad (82)$$

Denote by $\hat{\mathbb{E}}_n$ the empirical expectation of n i.i.d. random variables distributed by $p(t, \cdot)$. Write $\hat{\mathbb{E}}_n(\nabla F(X) \nabla F(X)^{\top}) = \mathbb{E}(\nabla F(X) \nabla F(X)^{\top}) + \delta_{1,n}/\sqrt{n}$ and $\hat{\mathbb{E}}_n \Delta F(X) = \mathbb{E} \Delta F(X) + \delta_{2,n}/\sqrt{n}$. Note that

$$\begin{aligned} \hat{\theta} &= -\hat{\mathbb{E}}_n(\nabla F(X) \nabla F(X)^{\top})^{-1} \hat{\mathbb{E}}_n \Delta F(X) \\ &= - \left(\underbrace{\mathbb{E}(\nabla F(X) \nabla F(X)^{\top})^{-1} \hat{\mathbb{E}}_n(\nabla F(X) \nabla F(X)^{\top})}_{I + \mathbb{E}(\nabla F(X) \nabla F(X)^{\top})^{-1} \delta_{1,n}/\sqrt{n}} \right)^{-1} \\ &\quad \underbrace{\mathbb{E}(\nabla F(X) \nabla F(X)^{\top})^{-1} \hat{\mathbb{E}}_n \Delta F(X)}_{-\theta_* + \mathbb{E}(\nabla F(X) \nabla F(X)^{\top})^{-1} \delta_{2,n}/\sqrt{n}} \\ &= \left(-I + \mathbb{E}(\nabla F(X) \nabla F(X)^{\top})^{-1} \frac{\delta_{1,n}}{\sqrt{n}} + \mathcal{O}\left(\frac{1}{n}\right) \right) \\ &\quad \left(-\theta_* + \mathbb{E}(\nabla F(X) \nabla F(X)^{\top})^{-1} \frac{\delta_{2,n}}{\sqrt{n}} \right) \\ &= \theta_* - \mathbb{E}(\nabla F(X) \nabla F(X)^{\top})^{-1} \frac{\theta_* \delta_{1,n} + \delta_{2,n}}{\sqrt{n}} + \mathcal{O}(1/\sqrt{n}), \end{aligned}$$

where we use Slutsky's theorem in the third equation. Moreover,

$$\begin{aligned} & \frac{\theta_* \delta_{1,n} + \delta_{2,n}}{\sqrt{n}} \\ &= \mathbb{E}(\nabla F(X) \nabla F(X)^\top \theta_* + \Delta F(X)) - \widehat{\mathbb{E}}_n(\nabla F(X) \nabla F(X)^\top \theta_* + \Delta F(X)), \end{aligned}$$

which converges in distribution to $\mathcal{N}(0, \Sigma_{\nabla F(X) \nabla F(X)^\top \theta_* + \Delta F(X)})$. Combining the above estimates yields (81). $\square$

Under the assumptions in Theorem 7.1, it is clear that Assumption 6.1 (L^2 -bound on score matching) holds if

$$\mathbb{E}_{p(t, \cdot)} \left(\sup_{\theta} |\partial_{\theta} p_{\theta}(t, X)|^{2+\epsilon} \right) < \infty,$$

for any $\epsilon > 0$. Another interesting result from [45] shows that under an isoperimetric assumption on $p(t, \cdot)$, the statistical efficiency of score matching is close to the maximum likelihood estimate by comparing Γ_{SM} with $\Gamma_{\text{MLE}} = \Sigma_{F(X)}^{-1}$. Similar results were obtained by [65] for a more general class of distributions.

7.2. Score matching by neural nets

Here we consider score matching for VP. Recall $\gamma(t) := \exp\left(-2 \int_0^t \beta(s) ds\right)$. By Tweedie's formula [18, 72], the score function can be written as:

$$\nabla \log p(t, x) = \frac{\gamma(t)^{\frac{1}{4}}}{1 - \sqrt{\gamma(t)}} \mathbb{E}(X_0 | X_t = x) - \frac{1}{1 - \sqrt{\gamma(t)}} x, \quad (83)$$

Inspired by the special structure (83), it is natural to take the function approximation:

$$s_{\theta}(t, x) = \frac{\gamma(t)^{\frac{1}{4}}}{1 - \sqrt{\gamma(t)}} f_{\theta}(t, x) - \frac{1}{1 - \sqrt{\gamma(t)}} x, \quad (84)$$

with $f_{\theta}(t, x)$ a suitable neural net (e.g. ReLU net). As pointed out by [7], the specific form (84) has close affinity to the U-net, where the term $-\frac{1}{1 - \sqrt{\gamma(t)}} x$ corresponds to a shortcut connection.

The following theorem provides a template for a class of results concerning the approximation efficiency of score matching. To simplify the presentation, we take the weight $\lambda(t) = 1$.

Theorem 7.2. [7, 29] *Let $\widehat{s}_{\theta}(\cdot, \cdot)$ solve one of the empirical score matching problems (in Section 4), where $\mathbb{E}$ is replaced with $\widehat{\mathbb{E}}$ under finite samples. Let $\widehat{s}_{\theta, k}$ be the output at iteration k of some algorithm (e.g. SGD). For $\widehat{s} = \widehat{s}_{\theta}$ or*

$\hat{s}_{\theta,k}$, we have:

$$\begin{aligned} & \int_0^T \mathbb{E}_{p(t,\cdot)} |\hat{s}(t,\cdot) - \nabla \log p(t,\cdot)|^2 dt \\ & \leq \int_0^T \mathbb{E}_{p(t,\cdot)} |\hat{s}(t,\cdot) - \nabla \log p(t,\cdot)|^2 - \widehat{\mathbb{E}}_{p(t,\cdot)} |\hat{s}(t,\cdot) - \nabla \log p(t,\cdot)|^2 dt \quad (85) \\ & \quad + \inf_{\theta} \int_0^T \widehat{\mathbb{E}}_{p(t,\cdot)} |s_{\theta}(t,\cdot) - \nabla \log p(t,\cdot)|^2 dt, \end{aligned}$$

where the first term on the right side of (85) is the sample error of score matching, and the second term is the approximation error.

In [7], $f_{\theta}(t, x)$ is taken to be a multilayer ReLU with linear structure. But no specific algorithm of stochastic optimization is used (corresponding to $\hat{s} = \hat{s}_{\theta}$). [29] considered a two-layer ReLU net, and gradient descent is used to update $\hat{s} = \hat{s}_{\theta,k}$. In particular, they used a neural tangent kernel as a universal proxy to the score, and showed that score matching (using a two-layer ReLU net) is as easy as kernel regression. [94] considered a feedforward net, also with gradient descent to update $\hat{s}_{\theta,k}$. In general, the bound (85) depends, in a complicated manner, on the data distribution, the net structure, the sample size, and the number of iterations in the optimization algorithm.

7.3. Score matching by reinforcement learning

In this last subsection, we consider score matching by RL, which was initiated by [4, 20, 49]. The idea is to treat the score as the action in RL, where the reward measures how well the sample Y_T aligns with human preferences (diffusion model alignment). Since the diffusion model is specified by the SDE, the recently developed continuous RL [38, 39, 93, 103] provides a convenient framework to formulate score matching by RL.

Let's recall the setup of continuous RL. The state dynamics $(X_t^a, t \geq 0)$ is governed by the SDE:

$$dX_t^a = b(t, X_t^a, a_t) dt + \sigma(t) dB_t, \quad X_0^a \sim p_{\text{init}}(\cdot), \quad (86)$$

where $b : [0, T] \times \mathbb{R}^d \times \mathcal{A} \rightarrow \mathbb{R}^d$, $\sigma : [0, T] \rightarrow \mathbb{R}_+$, and the action $a_s \in \mathcal{A}$ is generated from a distribution $\pi(\cdot | t, X_t^a)$ by external randomization. Define the (state) value function given the feedback policy $\{\pi(\cdot | t, x) : x \in \mathbb{R}^d\}$ by

$$V(t, x; \pi) := \mathbb{E} \left[\int_t^T r(s, X_s^{\pi}, a_s^{\pi}) ds + h(X_T^{\pi}) \middle| X_0^{\pi} = x \right],$$

where $r : \mathbb{R}_+ \times \mathbb{R}^d \times \mathcal{A} \rightarrow \mathbb{R}$ and $h : \mathbb{R}^d \rightarrow \mathbb{R}$ are the running and terminal rewards, respectively. The performance metric is $\eta(\pi) := \int_{\mathbb{R}^d} V(0, x; \pi) p_{\text{init}}(dx)$. The main task of continuous RL is to approximate $\max_{\pi} \eta(\pi)$ by constructing a sequence of policies π_k , $k = 1, 2, \dots$ recursively such that $\eta(\pi_k)$ is non-decreasing.

Now we introduce the main idea from [101, 102] (see also [25]). Comparing the backward process (6) with the continuous RL (86), it is natural to set $p_{\text{init}}(\cdot) = p_{\text{noise}}(\cdot)$,

$$b(t, x, a) = -f(T - t, x) + g^2(T - t)a \quad \text{and} \quad \sigma(t) = g(T - t). \quad (87)$$

Here the action a has a clear meaning – it is at the place of the score function. So we can deploy the continuous RL machinery to “fine-tune” the score. At time t , we use a Gaussian exploration $a_t^\theta \sim \pi_\theta(\cdot | t, Y_t^\theta) = N(\mu_\theta(t, Y_t^\theta), \sigma_t^2 I)$, where σ_t is a predetermined exploration level for each t . Under the Gaussian exploration, the RL dynamics is:

$$dY_t^\theta = [-f(T - t, Y_t^\theta) + g^2(T - t)\mu_\theta(t, Y_t^\theta)] dt + g(T - t)dB_t, \quad Y_0^\theta \sim p_{\text{noise}}(\cdot). \quad (88)$$

Denote by $p_\theta(t, \cdot)$ the probability distribution of Y_t^θ . In particular, the score matching parametrization $\mu_\theta(t, Y_t^\theta) = s_\theta(T - t, Y_t^\theta)$ recovers the backward process (25).

In order to prevent the model from overfitting to the reward or catastrophic forgetting, we add a penalty term to ensure that the pretrained model $s_{\text{pre}}(\cdot, \cdot)$ is fully exploited. The problem that we want to solve is:

$$\max_{\theta} \mathbb{E} [R(Y_T^\theta) - \beta \text{KL}(p_\theta(T, \cdot), p_{\theta_{\text{pre}}}(T, \cdot))], \quad (89)$$

where $\beta > 0$ is a (given) penalty cost, $R : \mathbb{R}^d \rightarrow \mathbb{R}$ is a reward to evaluate the sample Y_T^θ . It is easy to see that the objective function (89) is equivalent to:

$$\mathbb{E} \int_0^T \underbrace{-\frac{\beta}{2} g^2(T - t) |a_t^\theta - \mu_{\theta_{\text{pre}}}(t, Y_t^\theta)|^2}_{r(t, Y_t^\theta, a_t^\theta)} dt + \underbrace{\mathbb{E} R(Y_T^\theta)}_{h(Y_T^\theta)},$$

We also define the corresponding value function:

$$V(t, x; \pi_\theta) = \mathbb{E} \left[\int_t^T -\frac{\beta}{2} g^2(T - t) |a_t^\theta - \mu_{\theta_{\text{pre}}}(t, Y_t^\theta)|^2 dt + R(Y_T^\theta) \mid Y_t^\theta = x \right]. \quad (90)$$

The following theorem gives the policy gradient with respect to the performance metric η .

Theorem 7.3 (Policy gradient for score matching). [25, 102] *The policy gradient of a policy π^θ parameterized by θ is:*

$$\nabla_\theta V(t, x; \pi_\theta) = \mathbb{E} \left[\int_t^T \nabla_\theta \log \pi_\theta(a_s^\theta | s, Y_s^\theta) q(t, Y_s^\theta, a_s^\theta; \pi_\theta) ds \mid Y_t^\theta = x \right], \quad (91)$$

where $q(t, x, a; \pi_\theta) = \frac{\partial V}{\partial t}(t, x; \pi_\theta) + g^2(T - t) \frac{\partial V}{\partial x}(t, x; \pi_\theta) a + r(t, x, a)$ is the advantage function in continuous RL.

Proof. It follows from [38, Theorem 5] that for all t, x ,

$$\begin{aligned} & \nabla_{\theta} V(t, x; \pi_{\theta}) \\ &= \mathbb{E} \left[\int_t^T \nabla_{\theta} \log \pi_{\theta}(a_s^{\theta} | s, Y_s^{\theta}) \left(dV(s, Y_s^{\theta}; \pi_{\theta}) + R(Y_s^{\theta}) ds \right) \middle| Y_t^{\theta} = x \right]. \end{aligned} \quad (92)$$

By applying Itô's formula to $V(t, Y_t^{\theta}; \pi_{\theta})$, we have:

$$\begin{aligned} & dV(t, Y_t^{\theta}; \pi_{\theta}) \\ &= \left[\partial_t V(t, Y_t^{\theta}; \pi_{\theta}) + \frac{1}{2} g^2(T-t) \Delta V(t, Y_t^{\theta}; \pi_{\theta}) - f(T-t, Y_t^{\theta}) \frac{\partial V}{\partial x}(t, Y_t^{\theta}; \pi_{\theta}) \right. \\ & \quad \left. + g^2(T-t) \frac{\partial V}{\partial x}(t, Y_t^{\theta}; \pi_{\theta}) a_t^{\theta} + r(t, Y_t^{\theta}, a_t^{\theta}) \right] dt + \frac{\partial V}{\partial x}(t, Y_t^{\theta}; \pi_{\theta}) g(T-t) dB_t. \end{aligned} \quad (93)$$

Injecting (93) into (92), and noting $g^2(T-t) \frac{\partial V}{\partial x}(t, Y_t^{\theta}; \pi_{\theta}) a_t^{\theta} + r(t, Y_t^{\theta}, a_t^{\theta})$ are the only terms involving a_t^{θ} yields the desired result. $\square$

Relying on Theorem 7.3, the machinery in [103] provides various algorithms (e.g., proximal policy optimization (PPO) [74]) to implement score fine-tuning.

8. Conclusion and future directions

In this section, we collect a few open problems which we hope to trigger future research.

1. The theory for the ODE samplers, and especially the consistency model is sparse. [52] studied the convergence rate of a modified CD. It would be useful to establish provable/sharp guarantees for the original CD.
2. There has also been work on diffusion models in constraint domains [17, 21, 59, 71]. Establishing the convergence result in each particular case is of independent interest.
3. Approximation efficiency for score matching were studied in simple scenarios, e.g., exponential families, two-layer neural nets. It would be interesting to see whether one can go beyond these cases to match the practice.
4. Diffusion guidance [16, 32] and fine-tuning [90, 91] are used to generate data that is customized for specific downstream tasks. Some recent work [28, 86, 88, 96] explored the theoretical aspects of these approaches, which calls for further development.
5. Diffusion model alignment is a promising subject in the context of RL from human feedback (RLHF). Rudiments are provided in Section 7.3, and it remains to build an entire framework, with provable guarantees.

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